

PD-1 negatively regulates helper T cell differentiation into Th2

Reviewed Preprint

v1 • October 8, 2024

Not revised

Masaki Tajima, Naoko Ikuta, Yuka Nakajima, Kensuke Suzuki, Yosuke Tokumaru, Peng Li, Hiroshi Kiyonari, Tasuku Honjo, Akio Ohta 

Department of Immunology, Institute of Biomedical Research and Innovation, Foundation for Biomedical Research and Innovation at Kobe, Kobe, Japan • Division of Integrated High-Order Regulatory Systems, Center for Cancer Immunotherapy and Immunobiology, Graduate School of Medicine, Kyoto University, Kyoto, Japan • Pharmaceutical R&D Division, Meiji Seika Pharma Co. Ltd., Tokyo, Japan • Laboratory for Animal Resources and Genetic Engineering, RIKEN Center for Biosystems Dynamics Research, Kobe, Japan • Department of Immunology and Genomic Medicine, Center for Cancer Immunotherapy and Immunobiology, Graduate School of Medicine, Kyoto University, Kyoto, Japan

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Programmed Cell Death Protein-1 (PD-1) represents endogenous mechanisms of negative immunoregulation. While the modulation of effector functions has been the major focus of PD-1 research, quick PD-1 upregulation in naïve T cells starting 1 h after priming raised a possibility that PD-1 also affects the development of effector T cells. The role of PD-1 in functional differentiation into Th1 and Th2 has been unclear. In murine naïve CD4⁺ T cell activation, we found that PD-1 stimulation during the early stage of T cell activation strongly impaired Th2 cell development, while Th1 cell induction was relatively resistant to this immunosuppressive signaling. The steep decline in Th2 cell induction suggested the significance of PD-1 in allergic inflammation. Treatment with anti-human PD-1 agonist antibody inhibited allergic inflammation in human PD-1-knock-in mice as shown by the reduction of Th2 cells, IgE levels and eosinophilic infiltration. This study shows that PD-1 regulates not only the intensity but also the quality of immune response by deviating Th differentiation. PD-1 stimulators are projected to be valuable in suppressing various forms of inflammatory activities, but the efficacy against Th2-dominant immune response may be particularly high.

eLife Assessment

This **valuable** study reports on a novel role of PD-1 in early T cell differentiation, showing that PD-1 stimulation impairs Th2 differentiation more effectively than that of Th1, with implications for the treatment of allergies. However, whereas the series of well-designed experiments using OVA-specific CD4 T cells from DO.11.10 mice and the use of an allergy model generated **compelling** data, the study is still **incomplete** since it shows gaps in the rationale for the experimental protocols, contradictory data regarding IFN-gamma and IL-4 production, and the lack of in vivo experiments on Th2 differentiation to further support the main hypothesis. Nonetheless, the reported data would be of interest to immunologists working on T cell differentiation and allergy.

<https://doi.org/10.7554/eLife.100879.1.sa2>

Introduction

Th2 cells, which are CD4⁺ T cells secreting IL-4, IL-5 and IL-13, drive allergic inflammation as a major component of type 2 immune response (1, 2). Overactive type 2 immune response is responsible for the typical allergic diseases involving IgE-mediated degranulation of mast cells. IL-4 promotes B cell response and antibody class switch to IgE. IL-13 was shown to cooperate with IL-4 to produce high-affinity IgE, which is superior in stimulating mast cells than low-affinity IgE (3). IL-5 and IL-13 can promote allergic inflammation in an IgE-independent manner by activating eosinophils. These cytokines may be also produced from other effectors of allergic inflammation such as innate lymphoid cells (ILC) 2, mast cells and basophils, and mediate a positive feedback loop expanding type 2 immunity (4, 5). The growing prevalence of allergic diseases has become a worldwide major health concern. The prevention of overactive type 2 immune response is crucial to the control and treatment of various allergic disorders.

Recent advances in the understandings of type 2 immunity revealed that early production of IL-25, IL-33 and thymic stromal lymphopoietin (TSLP) from stressed epithelium can initiate type 2 immune responses (2, 6). ILC2 respond to these epithelial cell-derived cytokines and translate the danger signal into setting up the cytokine milieu that can specifically promote functional differentiation of naïve CD4⁺ T cells to Th2. However, other factors such as T cell receptor (TCR) signal intensity or co-stimulation can also affect Th2 cell differentiation (1, 7), and their relative importance remains to be elucidated (8, 9).

PD-1 represents negative co-stimulatory receptors and is indispensable in controlling T cell activities. The engagement of PD-1 with its natural ligands, PD-L1 or PD-L2, recruits Src homology-2 domain-containing protein tyrosine phosphatase-2 (SHP2), which in turn dephosphorylates the components of TCR signaling cascade and attenuates T cell activation (10–12). The significance of PD-1-dependent regulation has been evident from the exaggerated inflammatory activities in mice lacking PD-1 or PD-L1 (12). This discovery led to the emergence of PD-1 as a target of immune manipulation, and blockers of PD-1 pathway have seen a clinical success in cancer immunotherapy (13–15). Of note, eosinophilia and adverse allergic inflammation have been reported in some cancer patients after the treatment with PD-1 blocking antibodies (16–19). These symptoms imply the relevance of PD-1 pathway in modulating Th2 cells and type 2 immune response. The association of PD-1 and allergic inflammation has been studied in animal models using blocking antibodies or knockout mice of PD-1 or PD-L1; however, the results are surprisingly inconsistent. In allergic conjunctivitis and dermatitis models, the blockade of PD-1

or PD-L1 increased Th2 cells but failed to enhance inflammation (20 [↗](#), 21 [↗](#)). For the allergic asthma induction, the blockade of PD-1 or PD-L1 did not exaggerate airway hypersensitivity or eosinophils infiltration (22 [↗](#), 23 [↗](#)), and asthma induction in PD-L1-knockout mice rather reduced lung inflammation (24 [↗](#)).

In this study, our purpose is to elucidate the role of PD-1 in Th2 cell development. PD-1 expression is not found on naïve T cells but is upregulated in activated T cells. Therefore, most PD-1 studies have focused on already established effector T cells. However, our finding of PD-1 upregulation in the early phase of T cell activation prompted us to examine the impact of PD-1 stimulation on the direction of effector T cell development. Recently, we established PD-1 agonist antibodies, which can trigger the immunosuppressive activity of PD-1 and suppress inflammatory diseases in vivo (25 [↗](#)). PD-1 stimulation during the functional differentiation of naïve CD4⁺ T cells was found to strongly impair the induction of IL-4-producing Th2 cells, suggesting the feasibility of PD-1 as a target of therapeutic intervention to Th2-related diseases.

Results

Early PD-1 signaling substantially affects Th1/Th2 balance

PD-1 expression is not detectable in naïve T cells, but its upregulation after activation serves as a negative feedback mechanism to limit effector T cell activities. PD-1-dependent immunoregulation has been shown to critically affect anti-pathogen response and autoimmunity by activated T cells (12 [↗](#)). Besides the control of effector T cell activities, we hypothesized that PD-1 stimulation may change the course of effector T cell development when taking place in the early phase of T cell activation. To examine PD-1 induction during T cell activation, naïve CD4⁺ T cells from MHC class II-restricted TCR-transgenic DO11.10 mice were stimulated with OVA₃₂₃₋₃₃₉ peptide. T cells rapidly downregulated CD62L and started to upregulate PD-1 within 1 h post stimulation (Figure 1A, B [↗](#)). PD-1 levels progressively increased thereafter. The extent of PD-1 induction was associated with the antigen dose, and PD-L1 expression on antigen-presenting cells (APC) did not affect PD-1 induction (Figure 1C [↗](#)). PD-L1 on APC was suppressive to the freshly activated DO11.10 cells since the blockade of PD-L1 enhanced cumulative cell proliferation and cytokine production during first three days (Figure 1D, E [↗](#)). It should be noted that the optimal PD-1 upregulation was evident before the stimulated T cells started proliferation (Figure 1C, D [↗](#)). T cells upregulated PD-1 as soon as receiving TCR stimulation, raising a possibility that PD-1 signaling in the early stage of T cell activation can affect the direction of effector T cell development.

Upon stimulation, naïve CD4⁺ T cells differentiate into functionally distinct helper subsets: Th1, Th2, Th17 or regulatory T cells among others (26 [↗](#)). The intensity of T cell stimulatory signaling, which is affected by TCR affinity to the MHC complex together with numbers and duration of TCR-MHC interaction, can considerably affect the direction of naïve CD4⁺ T cell differentiation (26 [↗](#), 27 [↗](#)). Since PD-1 stimulation disrupts TCR signaling cascade, the early PD-1 induction during T cell activation may influence functional differentiation of CD4⁺ T cells. T-bet and GATA3 are master transcription factors of differentiation and functions of Th1 and Th2 cells, respectively. Naïve DO11.10 CD4⁺ T cells undergoing functional differentiation into Th1 or Th2 upregulate either transcription factor, but the expression of T-bet and GATA3 in CD4⁺ T cells is little on day 1 compared to the expression by vast majority on day 4 (Figure 2A [↗](#)). In contrast, PD-1 induction was evident within hours, well before the determination for the direction of CD4⁺ T cell differentiation into Th1 or Th2.

To examine the direct impact of PD-1 stimulation during the early phase of functional differentiation, DO11.10 naïve CD4⁺ T cells were primed by PD-L1-expressing or PD-L1-deficient APC. After 3 days, the same number of activated DO11.10 cells were restimulated with immobilized anti-CD3 mAb for the evaluation of cytokine-producing activity. CD4⁺ T cell activation

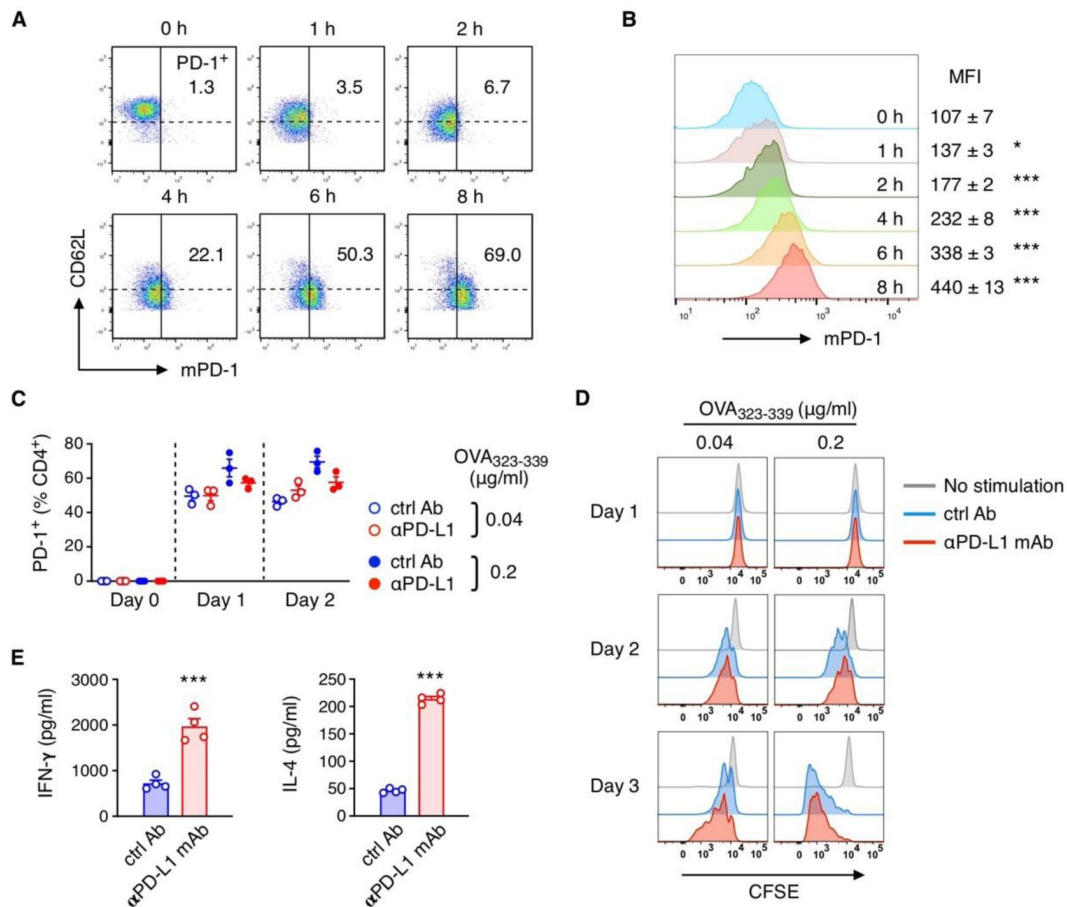


Figure 1.

Early PD-1 upregulation in activating T cells.

(A, B) PD-1 upregulation after antigenic stimulation of CD4⁺ T cells. Purified CD4⁺ CD62L⁺ T cells from DO11.10 TCR-transgenic mouse were stimulated with OVA₃₂₃₋₃₃₉ peptide (1 µg/ml) in the presence of A20 cells for indicated duration of time. Numbers represent percentages of PD-1⁺ within CD4⁺ cells (A) and mean fluorescence intensity of PD-1 (B). Data represent average ± SEM of triplicate samples. (C) PD-1 upregulation on DO11.10 T cells by different concentrations of OVA peptide. Anti-PD-L1 mAb or isotype control (5 µg/ml) was added at the beginning of co-culture. (D) T cell proliferation after stimulation. CFSE-labeled DO11.10 CD4⁺ CD62L⁺ T cells were stimulated as in (C). (E) PD-1-dependent inhibition of cytokine production from DO11.10 T cells. Cytokine levels in the culture supernatant were determined after 3 days of stimulation with OVA₃₂₃₋₃₃₉ (0.2 µg/ml). Data represent average ± SEM of triplicate (B, C) or quadruplicate (E) samples. *p < 0.05, ***p < 0.001; Student's t-test.

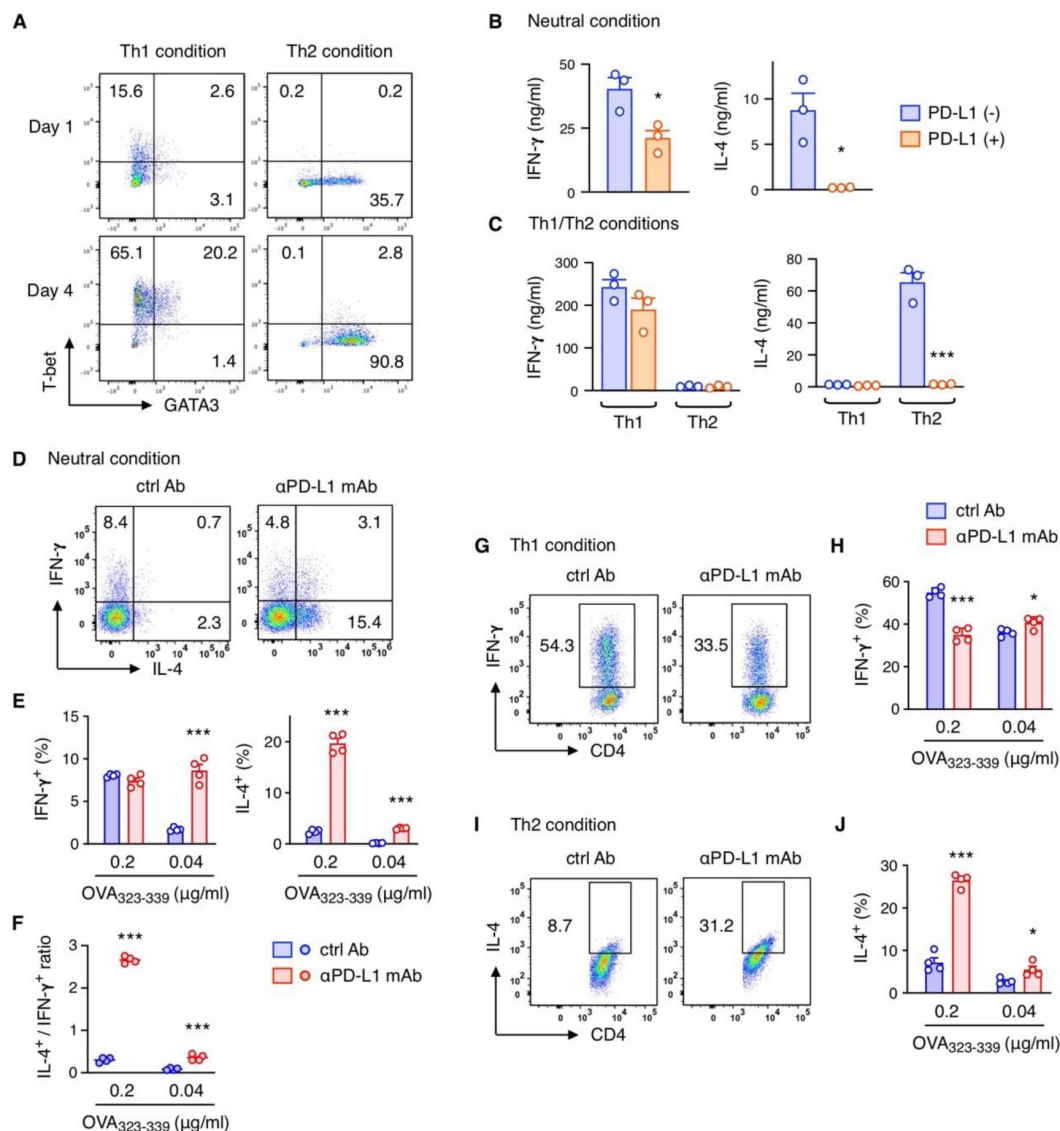


Figure 2.

PD-L1 blockade unleashes the development of IL-4-producing Th2 cells.

(A) Development of effector cells expressing T-bet or GATA3. Purified CD4⁺ CD62L⁺ DO11.10 T cells were stimulated with OVA₃₂₃₋₃₃₉ (1 μ g/ml) in a Th1 or Th2-skewing condition. Intracellular staining for T-bet and GATA3 expression was performed after 1 and 4 days. (B, C) PD-1-dependent regulation of Th1 and Th2 cell differentiation. Naïve DO11.10 T cells were stimulated in the neutral condition (B) or Th1/Th2-skewing conditions (C) for 3 days using PD-L1-deficient or PD-L1-expressing parental A20 cells as APC. Subsequently, cytokine-producing activities were evaluated by stimulating activated T cells (1.5×10^5 cells) with immobilized anti-CD3 mAb for 24 h. (D-J) Naïve DO11.10 T cells were stimulated for 3 days using parental A20 cells as APC in the presence or absence of anti-PD-L1 mAb. T cell stimulation was conducted in the neutral (D, F) or skewing condition for Th1 (G, H) and Th2 cells (I, J). Activated T cells were restimulated in the same condition for 2 more days. Intracellular staining for cytokines was performed on day 5. Percentages of IFN- γ - and IL-4- positive events in CD4⁺ cells (D, E, G-J) and IL-4⁺ event / IFN- γ ⁺ event ratios in individual samples (F) are shown. Data represent average $\pm$ SEM of triplicate (B, C) or quadruplicate (E, F, H, J) samples. * $p < 0.05$, *** $p < 0.001$; Student's t-test.

in the presence of PD-1 stimulation induced effector T cells that produce impaired levels of cytokines. Of note, IL-4-producing cells were totally disappeared, while 50% of IFN- γ production was still present (**Figure 2B**). This result is not due to the inhibition of effector function but represents the impaired establishment of cytokine-competent effector T cells because PD-L1 was present only at the priming phase. The greater degree of inhibition for IL-4-producing cells than IFN- γ -producers suggests that PD-1 stimulation during the priming phase critically affects Th2 cell development. When DO11.10 cells were skewed into Th1 or Th2 in the specific cytokine environments, the priming with PD-L1-expressing APC strongly impaired the development of IL-4-producing Th2 cells, but not IFN- γ -producing Th1 cells (**Figure 2C**).

The experiment above used two different cell lines, PD-L1-expressing and PD-L1-deficient B lymphoma, as APC. To exclude the possibility that different intensity of T cell stimulation by these two lines might have influenced the Th1 vs Th2 differentiation, we further compared CD4⁺ differentiation by PD-L1-expressing APC in the presence or absence of PD-L1 blocking antibody. DO11.10 T cell stimulation in the “neutral” condition where no cytokine milieu was manipulated except for IL-2 supplementation resulted in preferential Th1 induction. However, IL-4-producing Th2 cells prominently arose by PD-L1 blockade (**Figure 2D, E**). Consequently, Th1/Th2 ratio significantly shifted toward Th2 by PD-L1 blockade, suggesting that PD-1 stimulation is strongly suppressive to Th2 cell development over Th1 (**Figure 2F**). The same trend was observed in the enforced Th1- and Th2-skewing conditions. The significant increase of IL-4-producing cells by PD-L1 blockade indicated that early PD-1 signaling was strongly suppressive to Th2 cell induction, while Th1 development remained intact (**Figure 2G-J**). PD-1-PD-L1 interaction was suppressive to the proliferation of both Th1 and Th2 cells, suggesting that early PD-1 stimulation made a qualitative difference in the functional differentiation of CD4⁺ T cells.

PD-1 agonist antibody suppresses cytokine-producing Th2 cell induction

PD-1-PD-L1 interaction during functional differentiation of T cells was inhibitory to the subsequent development of Th2-type effectors. We next examined whether pharmacological PD-1 stimulatory agent could negatively modulate Th2 responses. The clinical success of immunoenhancing anti-PD-1 blocking antibodies in cancer treatment has indicated that PD-1 is a proven target of pharmacological modulation of the human immune system (15). To establish a new class of immunosuppressant, we clarified the specific requirements to anti-PD-1 agonist antibodies that can trigger the immunosuppressive activity of PD-1 (25). One of the most effective anti-human PD-1 (hPD-1) agonist antibodies, HM266, was used to stimulate PD-1 in the course of CD4⁺ T cell differentiation.

To utilize anti-hPD-1 agonist antibodies in the antigen-specific activation of mouse CD4⁺ T cells, hPD-1 gene was retrovirally transduced into DO11.10 CD4⁺ T cells. Besides normal hPD-1 (hPD-1 WT), we also used mutant hPD-1 that is unable to transduce intracellular signaling due to the tyrosine to phenylalanine mutations in the ITIM and ITSM motifs in the cytoplasmic domain (hPD-1 YFYF) (10, 11). DO11.10 cells expressing either hPD-1 WT or hPD-1 YFYF at equivalent levels were skewed into Th1 or Th2 cells in the specific cytokine environment (**Figure 3A**). HM266 strongly inhibited the establishment of IL-4-producers from hPD-1 WT-expressing cells in Th2-inducing condition, whereas the development of IFN- γ -producing Th1 cells remained intact (**Figure 3B, C**). The inhibitory effect of PD-1 agonists on Th2 development was more potent when CD4⁺ T cells were stimulated at a low antigen dose than a high dose. The inverse correlation between antigen dose and Th2 inhibition suggests that the balance between the stimulatory TCR signaling and the inhibitory PD-1 signaling during early T cell activation is critical to the function of resulted effector cells. HM266 did not affect the differentiation of hPD-1 YFYF-expressing T cells, confirming that the inhibitory effect of HM266 was mediated by PD-1 signaling (**Figure 3D-G**).

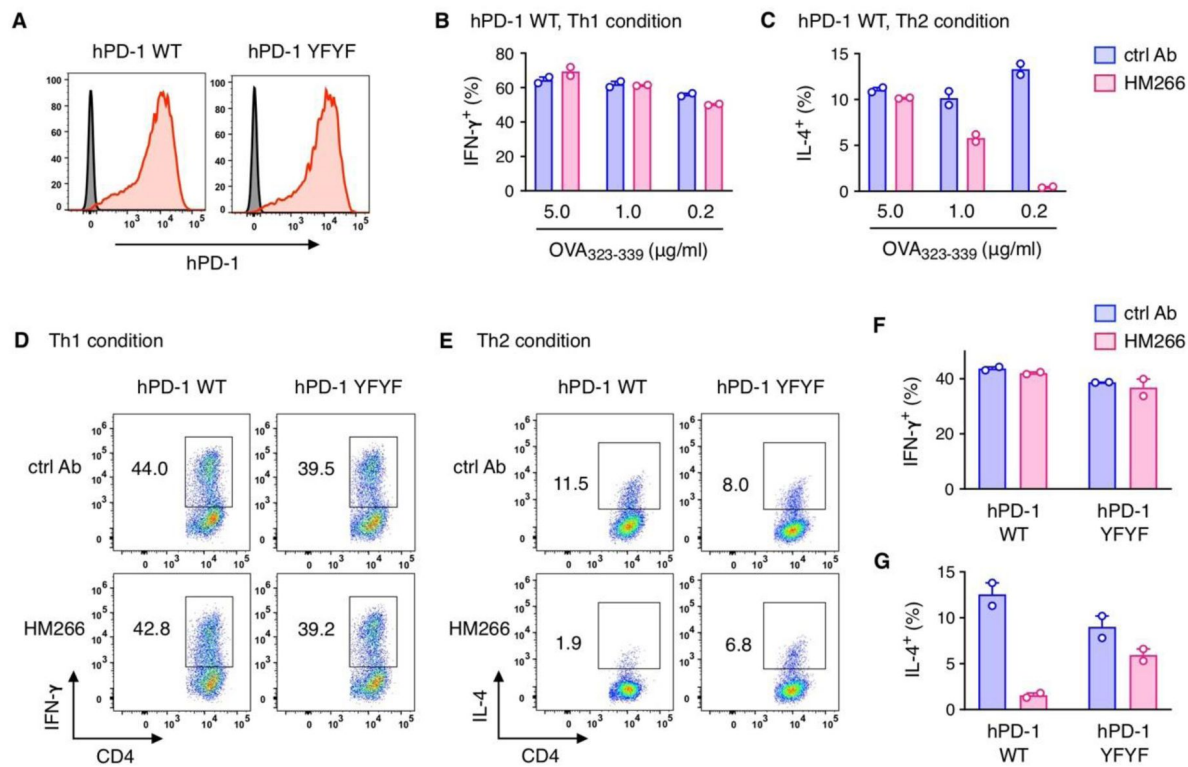


Figure 3.

PD-1 agonist antibody strongly inhibits the induction of cytokine-producing Th2 cells.

hPD-1-transduced DO11.10 T cells were stimulated with OVA₃₂₃₋₃₃₉ in Th1 or Th2-skewing condition using PD-L1-deficient A20 cells as APC. (A) hPD-1 expression in DO11.10 T cells after the transduction of wild-type hPD-1 or YFYF mutant. (B, C) The effects of anti-hPD-1 agonist mAb, HM266, on hPD-1 WT-transduced DO11.10 T cells in Th1 (B) or Th2-skewing condition (C). The concentrations of HM266 or isotype control were 5 µg/ml. Intracellular cytokine staining was performed on day 5. (D-G) The involvement of PD-1 signaling in Th2 inhibition by HM266. DO11.10 T cells expressing hPD-1 WT or hPD-1 YFYF were stimulated with OVA₃₂₃₋₃₃₉ (1 µg/ml) in Th1 or Th2-skewing condition. Representative plots of intracellular cytokine staining on day 5 (D, E) and proportions of IFN-γ⁺ (F) and IL-4⁺-producing cells (G) within CD4⁺ cells. Data represent average ± SEM of duplicate samples.

The attenuated IL-4-producing Th2 cell development by early PD-1 stimulation was further confirmed by another anti-hPD-1 agonist antibody, J116, which has the less potent agonistic activity than HM266 (**Figure 4A**). J116 still impaired the induction of IL-4-producing cells, but not IFN- γ -producers, as shown by intracellular staining (**Figure 4B-E**) and by cytokine production in restimulated CD4⁺ T cells (**Figure 4F, G**). Again, the defect of PD-1 signaling in hPD-1 YFYF-expressing cells abolished the inhibitory effect of J116 on Th2 development.

Next, we examined more physiologically-relevant condition, which allows development of both Th1 and Th2 cells, instead of the skewing conditions inducing exclusively Th1 or Th2 cells. Sensitization of bone marrow-derived dendritic cells (BMDC) with house dust mite (HDM) extract has been known to induce preferential Th2-type immune response (28). The culture supernatant of HDM-treated BMDC was added to hPD-1-transduced DO11.10 cells. In this setting, similar proportions of IFN- γ - and IL-4-producers were observed in activated DO11.10 cells. HM266 was inhibitory to the development of both types of effectors, but the degree of inhibition was greater for IL-4-producers than IFN- γ -producers (**Figure 5**).

The above experiments used hPD-1-transduced DO11.10 cells to test anti-hPD-1 agonist mAbs; therefore, those T cells express PD-1 at higher levels than that can be seen in spontaneous expression. To further confirm the attenuated Th2 development by PD-1 agonist at physiological levels of PD-1 expression, we produced hPD-1-knock-in (hPD-1-KI) DO11.10 mice, which express hPD-1 in place of mouse PD-1. These CD4⁺ T cells upregulated hPD-1 soon after the antigenic stimulation (**Figure 6A, B**). Stimulation of hPD-1 with HM266 in the early phase of functional differentiation diminished the development of Th2 cells, but not Th1 induction (**Figure 6C, D**). Cytokine levels from the restimulated T cells confirmed that T cell activation in the presence of PD-1 agonist mAb largely reduced IL-4/IL-13-producing Th2 cells, while IFN- γ -producing Th1 cells received only minor effect (**Figure 6E-G**). Taken together, PD-1 stimulation is inhibitory to the development of effector T cells, and its influence is substantial to Th2 immunity.

PD-1 agonist antibody suppresses allergic inflammation in vivo

Remarkable suppression of Th2 cell induction suggests that PD-1 is a potential target for the treatment of allergic diseases. To induce allergic asthma, we sensitized hPD-1-KI C57BL/6 mice with HDM extract. HDM extract is a potent allergen causing asthma in human and mice by the action of Th2-type cytokines such as IL-4, IL-5 and IL-13. Daily intranasal HDM challenge to the sensitized hPD-1-KI mice induced allergic inflammation as shown by massive infiltration of inflammatory cells in the lung. Eosinophils accounted for a large part of cell infiltrates in the bronchoalveolar lavage (BAL). CD4⁺ T cells in the inflamed lung and BAL expressed PD-1 along with CD69 and TIGIT (**Figure 7**, **Figure 7-supplement 1**).

The efficacy of HM266 against allergic inflammation was examined in HDM-induced asthma model. When HM266 treatment started at the time of sensitization (**Figure 7A**, preventive setting), the PD-1 agonist strongly reduced infiltration of eosinophils and CD4⁺ T cells in the BAL (**Figure 7B-D**) and mucus hyperproduction in the lung (**Figure 7E**). Considerable portion of CD4⁺ T cells in the BAL and lung tissue were Th2-type cells producing IL-4, IL-5 and/or IL-13. HM266 significantly reduced proportions of these cytokine-producing cells (**Figure 7F-H**). With the drop in CD4⁺ T cell number taken into calculation, the treatment with PD-1 agonist strongly prevented the bronchoalveolar Th2 cell accumulation (**Figure 7I**). Serum IgE levels were significantly decreased by HM266 corresponding to the reduction in Th2 cytokines (**Figure 7J, K**). Of note, PD-1 stimulation was effective even when applied only after the elicitation of lung inflammation (**Figure 7A**, therapeutic setting). Single dose of HM266 after four consecutive intranasal HDM challenges could significantly reduce bronchoalveolar infiltration of eosinophils and CD4⁺ T cells, especially producers of IL-5 and IL-13 (**Figure 7F-I**). IgE levels after the

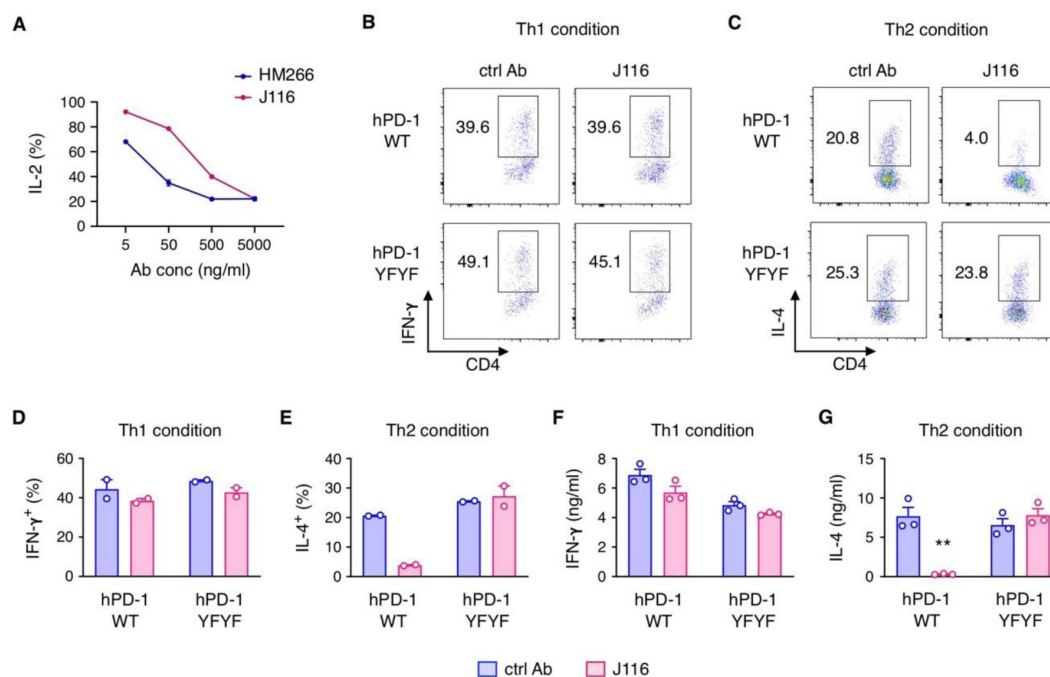


Figure 4.

J116 inhibits the induction of IL-4-producing Th2 cells.

(A) The immunosuppressive activity of anti-hPD-1 mAb J116. The PD-1 agonistic activity was determined by the inhibition of IL-2 production from DO11.10 T cell hybridoma. OVA₃₂₃₋₃₃₉ peptide (2 µg/ml) was added to stimulate hPD-1-transduced DO11.10 cells in the presence of IIA1.6 cells that express murine FcγRIIB but lack PD-L1. (B-G) DO11.10 T cells were transduced with wild-type hPD-1 or YFYF mutant and skewed into Th1 or Th2 cells in the presence of J116 or mouse IgG1 isotype control (5 µg/ml). The concentration of OVA₃₂₃₋₃₃₉ was 1 µg/ml. Intracellular cytokine staining was performed on day 5. Representative plots of cytokines staining (B, C) and percentages of IFN-γ- (D) and IL-4-producing cells (E) in CD4⁺ T cells. The same number of T cells were restimulated with anti-CD3 mAb on day 5, and IFN-γ- (F) and IL-4 levels (G) after 24 h were determined in the culture supernatant. Data represent average ± SEM of triplicate (A, F, G) or duplicate (D, E) samples. **p < 0.01; Student's t-test.

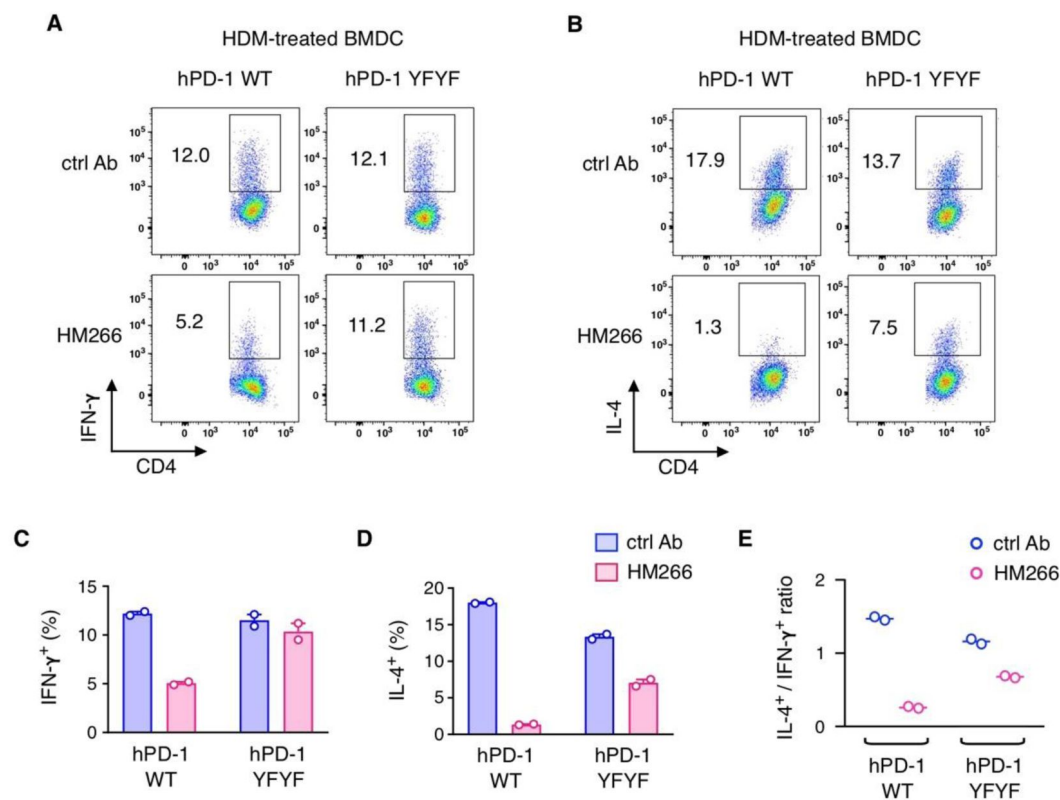


Figure 5.

Immune deviation by PD-1 agonist antibody in a Th2-prone condition.

hPD-1- transduced DO11.10 T cells were stimulated with OVA₃₂₃₋₃₃₉ (1 µg/ml) in the presence of HDM-stimulated BMDC supernatant using PD-L1-deficient A20 cells as APC. The concentrations of HM266 or isotype control were 5 µg/ml. (A, B) Representative plots of IFN-γ- (A) and IL-4-producing CD4⁺ cells (B) on day 5. DO11.10 T cells received transduction of either wild-type hPD-1 or YFYF mutant. (C, D) Proportions of IFN-γ- (C) and IL-4-producing cells (D) within CD4⁺ cells. (E) IL-4⁺ event / IFN-γ⁺ event ratios in individual samples. Data represent average ± SEM of duplicate samples.

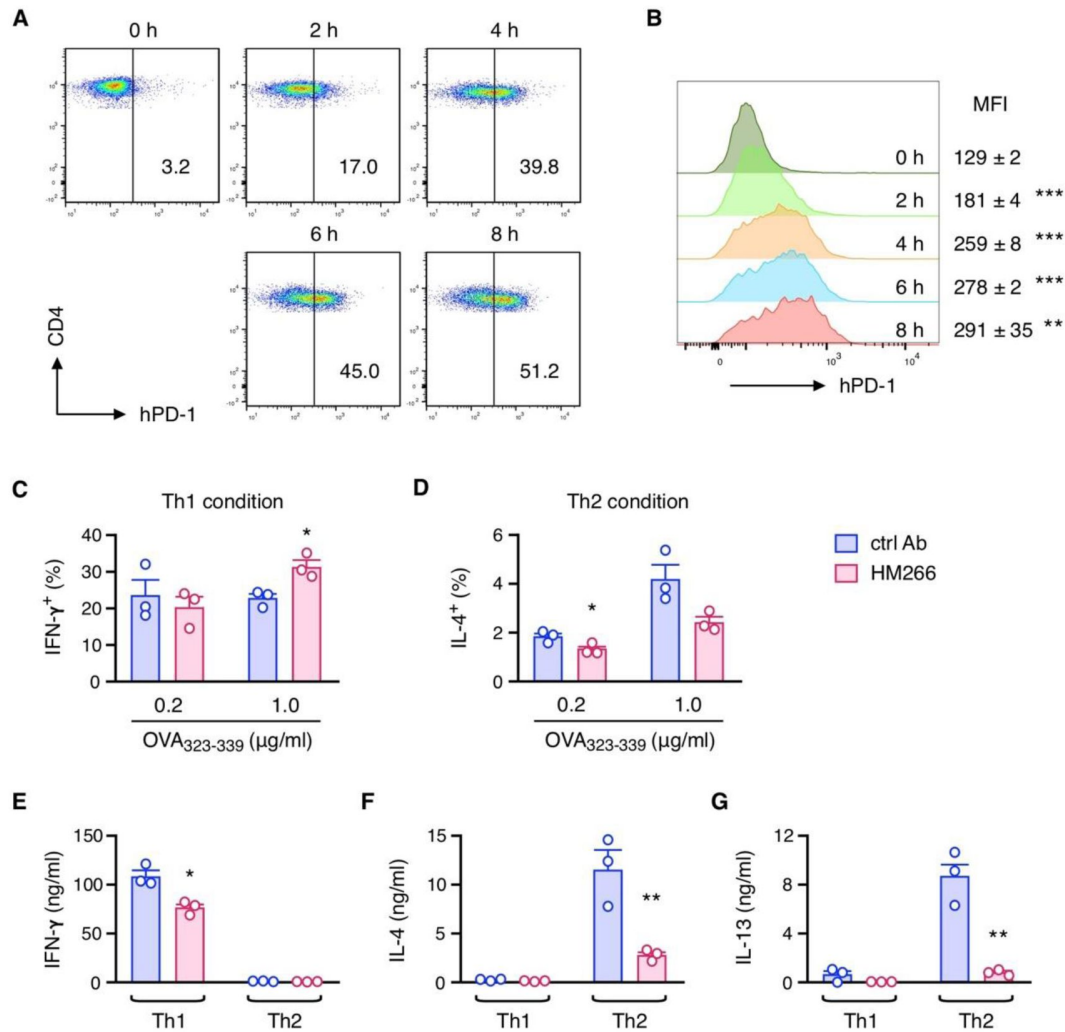


Figure 6.

Spontaneous PD-1 upregulation in T cells sufficiently suppress functional differentiation into Th2 cells when stimulated with PD-1 agonist antibody.

CD4⁺ CD62L⁺ cells from hPD-1-KI DO11.10 mice were stimulated with OVA₃₂₃₋₃₃₉-pulsed PD-L1-deficient A20 cells. (A, B) Early PD-1 upregulation in the stimulated hPD-1-KI DO11.10 cells. OVA₃₂₃₋₃₃₉ concentration was 1 μg/ml. Numbers indicate percentages of PD-1⁺ in CD4⁺ cells and mean fluorescence intensity of PD-1 (B). (C, D) The inhibition of Th2 cell induction by HM266, anti-hPD-1 agonist mAb. hPD-1-KI DO11.10 cells were stimulated with the indicated concentrations of OVA₃₂₃₋₃₃₉ for 3 days. HM266 or control mouse IgG1 was added at 5 μg/ml. The activated CD4⁺ T cells were restimulated with immobilized anti-CD3 mAb for 6 h. IFN-γ-producing cells in Th1-skewing condition (C) and IL-4-producing cells in Th2-skewing condition (D) were evaluated by subsequent intracellular cytokine staining. (E-G) Cytokine-producing activities of activated CD4⁺ T cells. IFN-γ (E), IL-4 (F) and IL-13 (G) levels in the culture supernatant were evaluated after restimulation of activated T cells (2 × 10⁵ cells) on day 3 with immobilized anti-CD3 mAb for 24 h. Data represent average ± SEM of triplicate samples. *p < 0.05, **p < 0.01, ***p < 0.001; Student's t-test.

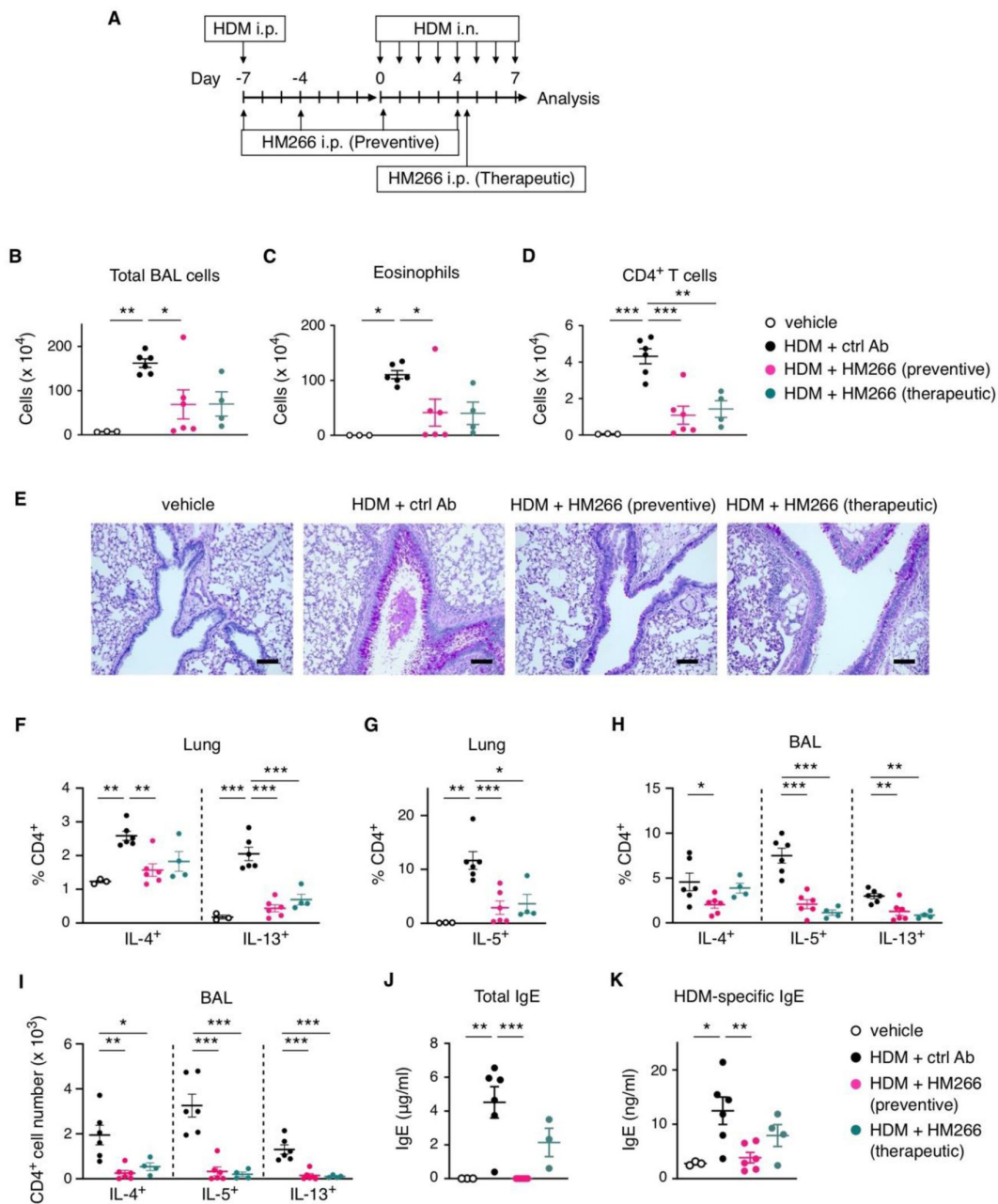


Figure 7.

PD-1 agonist antibody ameliorates allergic inflammation.

(A) The induction of allergic asthma in HDM-sensitized hPD-1 knock-in mice. Treatment with anti-hPD-1 agonist mAb, HM266, was done in two different schedules as indicated. (B-D) Infiltrated cell numbers in the BAL. Number of eosinophils (C) and CD4⁺ T cells (D) were calculated from flow cytometric analysis. (E) PAS staining of the lung sections. Scale bars represent 100 μ m. (F-H) Intracellular cytokine staining of CD4⁺ T cells from the lung tissue (F, G) and BAL fluid (H, I). Panels indicate percentages in CD4⁺ T cells (F-H) and the numbers of cytokine-producing CD4⁺ T cells. (J, K) Plasma levels of total IgE (J) and HDM-specific IgE levels (K) on day 7. Data represent average $\pm$ SEM of vehicle control (n = 3), HDM + ctrl Ab (n = 6), HDM + preventive HM266 (n = 6) and HDM + therapeutic HM266 (n = 4) groups. *p < 0.05, **p < 0.01, ***p < 0.001; Tukey-Kramer test.

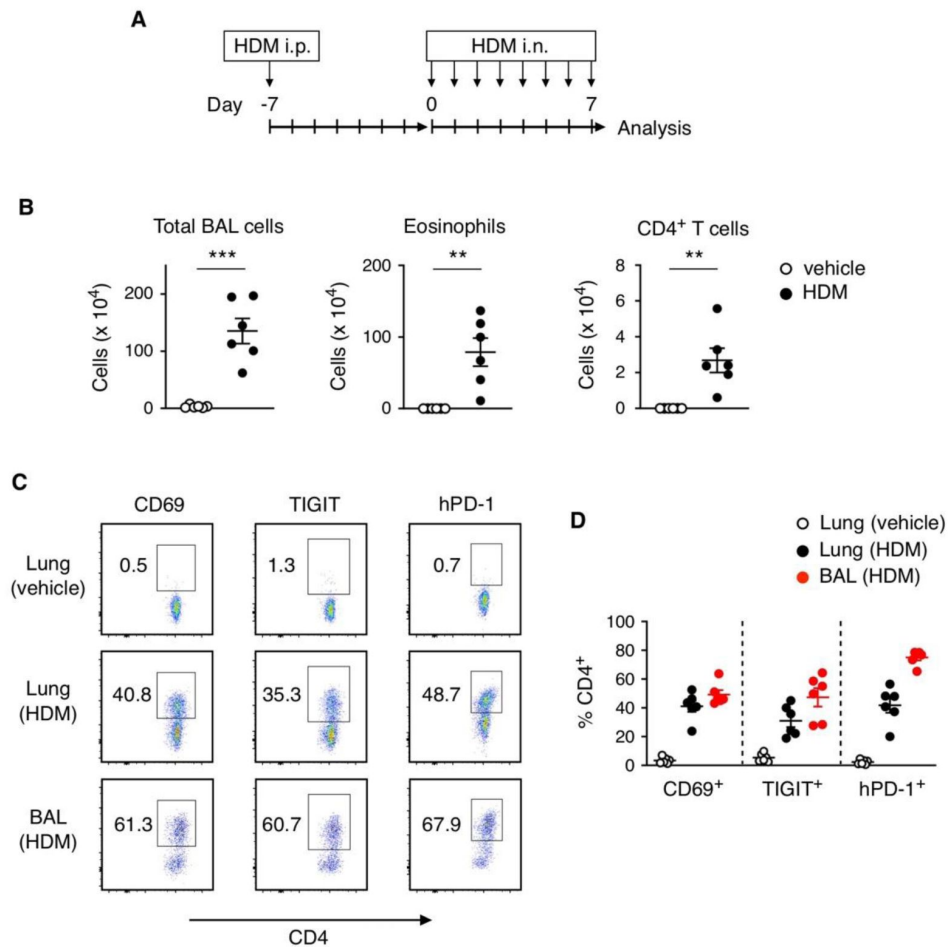


Figure 7, supplement 1.

PD-1 expression on CD4⁺ T cells from allergic hPD-1 knock-in mice.

(A) HDM extract induced lung allergy was induced on hPD-1 knock-in mice as indicated. Mice were euthanized 4 h after the final intranasal challenge. (B) BAL cell numbers. Numbers of eosinophils and CD4⁺ T cells were calculated from flow cytometry analysis. (C) Representative FACS plots of CD69, TIGIT, and hPD-1 expression on CD4⁺ T cells collected from BAL or lung cells. (D) Percentages of CD69⁺, TIGIT⁺, and hPD-1⁺ cells in CD4⁺ T cells. Data represent average $\pm$ SEM of 6 mice. ** $p < 0.01$, *** $p < 0.001$; Student's t-test.

therapeutic treatment with HM266 showed a decreasing trend, though it was not statistically significant. This result suggests that pharmacological PD-1 stimulation can alleviate allergic inflammation by blocking Th2-type immune response.

Discussion

In this study, antigenic stimulation of T cells was found to quickly induce PD-1 expression on their surface. PD-1 stimulation during early stage of effector T cell development inhibited the production of functional effector cells; however, this inhibitory activity did not equally affect functionally differentiated CD4⁺ T cell subsets. Specifically, the co-inhibitory signal of PD-1 severely impaired Th2 cell development. Such an immune bias by PD-1 stimulation was not observed in studies using gene-deficient mice.

Mice lacking PD-1 or PD-L1 indicated highly active Th1-type immune response in Leishmaniasis (29) and Schistosomiasis (30). Associated with the enhanced Th1 immunity, PD-1-knockout mice have shown intense CD8⁺ T cell-dependent proinflammatory activities in anti-tumor response (31–33) and in contact hypersensitivity (34, 35). In contrast, these gene-deficient mice failed to exaggerate allergic inflammation in allergic asthma (24) and atopic dermatitis models (21). The predominance of Th1-type immunity in PD-1/PD-L1-deficient mice might be stemmed from predisposed functional changes in immune cells. Indeed, treatment of wild-type mice with PD-1 blocking antibodies has provided different results from gene-deficient mice in the induction of allergic inflammation.

Experimentally, allergic diseases are often induced by sensitization of mice with an antigen and subsequent exposure of the target tissue with the same antigen. The blockade of PD-1 or PD-L1 during the challenge did not affect eosinophils activation or IgE increase (20, 22). However, PD-1/PD-L1 blockade during the sensitization phase significantly increased IL-4/IL-5/IL-13-producing cells in allergic conjunctivitis (20). In Schistosomiasis, PD-1 blockade was also shown to increase Th2 cells (36). Consistent with our study, these results using timely blockade approach suggested that PD-1 plays an important role in the regulation of Th2 cell induction.

Interestingly, the blockade of PD-L1 or PD-L2 has provided different results. While allergic inflammation did not change or rather attenuated in PD-L1-deficient mice (21, 24), PD-L2 deficiency exaggerated the induction of allergic asthma and dermatitis. Consistent with this difference, anti-PD-L2 blocking antibody induced stronger allergic inflammation than PD-L1 blockade (20, 22). The relative importance of PD-L2 in allergic inflammation may be due to its IL-4-inducible nature in contrast to IFN- γ -dependent induction of PD-L1 (37, 38). A complex regulation may be underlying with the PD-L2 involvement in allergic inflammation since PD-L2 was also suggested to affect Th1/Th2 balance in a PD-1-independent mechanism (23, 39).

To directly examine the role of PD-1 on Th functional differentiation, we compared DO11.10 T cell activation in the presence or absence of PD-1 stimulators. PD-1 stimulation is suppressive to the expansion of DO11.10 cells as previously reported (40), but CD4⁺ T cells that managed to proliferate in the presence of PD-1 stimulator were almost exclusively IFN- γ producers. PD-1 stimulation strongly impaired the development of IL-4-producing DO11.10 cells even in the Th2-skewing condition. Consistent with this view, cancer patients who have undergone PD-1 blockade for immunotherapy often experience eosinophilia (16, 17, 41–43), suggesting PD-1-dependent regulation of Th2 response in humans.

It remains to be elucidated why functional differentiation into Th2 is more sensitive to PD-1 stimulation than Th1 development. Extensive in vitro studies on Th differentiation have shown the importance of early IL-4 production in Th2 differentiation (44). The source of IL-4 has been investigated to identify the key player of Th2 induction. In T cells, the lack of IL-2 was shown to

impair Th2 differentiation, and early IL-2 production was suggested to mediate IL-4 induction (45 [↗](#), 46 [↗](#)). PD-1 stimulation with ligands or agonist antibodies downregulates IL-2 production from T cells (25 [↗](#), 47 [↗](#)). However, IL-2 downregulation was unlikely to account for the PD-1-dependent inhibition of Th2 differentiation because IL-2 supplementation did not reverse the impaired development of IL-4-producing cells even in the Th2-skewing condition.

Since the relation of TCR signal strength to Th differentiation has been suggested, PD-1-dependent attenuation of TCR signal strength might have affected Th2 differentiation. Stimulation of naïve Th cells with low-intensity TCR signals preferentially induced Th2 cells in experiments using small Ag dose and low-affinity antigenic peptides (46 [↗](#), 48 [↗](#)–50 [↗](#)). However, in different settings, low antigen dose could result in preferential Th1 induction both in vitro and in vivo (1 [↗](#), 51 [↗](#), 52 [↗](#)). In our study, the balance between immunostimulatory TCR signal and PD-1-dependent immunosuppressive signal significantly affected Th1/Th2 balance. PD-1 stimulation was previously reported to moderately modify the cytokine profile in favor of Th2 cells when T cells were stimulated with plate-bound anti-CD3 mAb and PD-L1-Fc (53 [↗](#)). Although it has been frequently used to study PD-1-mediated immunosuppression, the co-immobilization of PD-L1-Fc sometimes causes PD-1-independent immunosuppression most likely due to the reduction of T cell engagement with anti-CD3 mAb (25 [↗](#), 54 [↗](#)). The reason for the discrepancy with our result is unclear, but the difference in TCR signaling intensity might have influenced the outcome. Downstream TCR signaling, early ERK activation has been suggested to mediate Th2 differentiation (55 [↗](#)–57 [↗](#)). Inhibitory effect of PD-1 on ERK phosphorylation in the TCR signaling (25 [↗](#)) may be involved in the mechanism of the attenuated Th2 differentiation. Interestingly, CTLA-4-Ig was previously shown to inhibit Th2 induction (58 [↗](#)). The mechanism remains to be explored how co-stimulatory and co-inhibitory pathways influence the regulation of Th2 differentiation.

The overall role of PD-1 is suppressive to Th2 cells by inhibiting their establishment as well as effector functions, raising a possibility that PD-1 agonists are effective in the treatment of allergic diseases. We have previously identified anti-hPD-1 agonist antibodies that can stimulate the immunosuppressive activity of PD-1. The PD-1 agonist antibodies were effective in inhibiting various types of immune responses as shown in CD8⁺ T-dominant acute graft-versus-host disease and CD4⁺ T-dependent colitis induction (25 [↗](#)). Other group also showed the alleviation of ILC2-induced lung inflammation (59 [↗](#)) and neutrophilic lung inflammation by PD-1 agonist antibody (60 [↗](#)). In our current study, the PD-1 agonist antibodies demonstrated anti-inflammatory efficacy to eosinophilic inflammation in allergic asthma. The treatment prevented IgE increase, eosinophilic infiltration and mucus production, which are correlated with the suppression of Th2 cytokines. Importantly, PD-1 agonist was effective even after the elicitation of disease. Taken together, PD-1 agonist antibody holds a promise as a therapeutic of various allergic disorders through the inhibition of type 2 immunity by targeting both Th2 cells and ILC2.

In conclusion, we found that the co-inhibitory signal of PD-1 not only reduces the intensity of effector T cell activities but also deviates functional differentiation of CD4⁺ T cells. The stimulation of PD-1 signaling strongly inhibited Th2 induction, and PD-L1 blockade resulted in a permissive condition that allowed Th2 abundance. The significance of PD-1-dependent regulation in Th2 response has been also suggested in human patients who received anti-PD-1 blocking Ab for cancer immunotherapy. Eosinophilia in these patients may indicate the sign of immunoenhancement by PD-1 blockade just as concurrent immune-related adverse events (irAE) has been found to correlate with better prognosis (16 [↗](#), 17 [↗](#), 43 [↗](#)). Intense eosinophilic tissue inflammation sometimes complicates the anti-cancer treatment when PD-1 blockade induces severe allergic irAE such as drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome and acute eosinophilic pneumonia (18 [↗](#), 19 [↗](#)). PD-1 agonist Ab could regulate type 2 immunity to the opposite direction and prevent IgE increase and infiltration of eosinophils. The current study suggests the potential of PD-1-dependent immunoregulation as a target of anti-allergic treatment.

Materials and Methods

Mice

C57BL/6N mice were purchased from Japan SLC (Hamamatsu, Japan). DO11.10 TCR transgenic mice (Stock No: 003303) were purchased from the Jackson laboratory (Bar Harbor, ME). C57BL/6-background hPD-1-knock-in mice (Accession No. CDB0117E: <https://large.riken.jp/distribution/mutant-list.html>) were generated as described previously (25). hPD-1 knock-in DO11.10 mice were produced by breeding DO11.10 mice with BALB/c-background hPD-1-knock-in mice. BALB/c hPD-1-knock-in mice (Accession No. CDB0119E: <https://large.riken.jp/distribution/mutant-list.html>) were generated by the insertion of hPD-1 cDNA-bovine poly A signal into the start codon of mouse PD-1 genome in BALB/c embryos as described for C57BL/6-background hPD-1-knock-in mice (25). For the homologous recombination-mediated knock-in, the donor vector consisting of homology arms and hPD-1-bpA (bovine growth hormone poly A signal sequence) was generated to insert the hPD-1-bpA cassette at the 4 base upstream of the PAM sequence. gRNA site: 5'-GCC AGG GGC TCT GGG CAT GT-3'. Primer pairs for genotyping: wild-type FW 5'-AGG AGA CTG CTA CTG AAG GC -3' and REV 5'-CCA ATC CGT GTA ACC AGG -3' (245 bp); knock-in FW 5'-CAG GCC TCG ACA CCC ACC -3' and REV 5'-CAG CCC AGT TGT AGC ACC -3' (536 bp). All mice were maintained under specific pathogen free condition in the animal facility at Oriental BioService or RIKEN Kobe Branch. Both sexes were used for experiments. All animal experiment protocols were approved by the IACUC of Foundation for Biomedical Research Institute at Kobe (#20-04) and of RIKEN Kobe Branch (#A2017-08).

Cell lines

A20 cells (RRID: CVCL_1940) were provided by Dr. Tasuku Honjo (Kyoto University). PD-L1-deficient A20 cells were produced by the following method. Guide RNA (target sequence: TCCAAAGGACTTGTACGTGG) was prepared using the GeneArt Precision gRNA Synthesis Kit (Invitrogen, catalog# A29377) and introduced into A20 cells together with GeneArt Platinum Cas9 Nuclease (Invitrogen, catalog# B25640) using the Neon Transfection System (Invitrogen). PD-L1-negative cells were isolated by several rounds of FACS sorting, followed by limiting dilution. Clone with the similar capability to stimulate T cells as parental A20 cell was chosen and used for further studies. PD-1 knockout DO11.10 T cell hybridoma and PD-L1 knockout IIA1.6 cells were obtained from Dr. Taku Okazaki (Tokyo University). Cell culture was done using RPMI 1640 (Gibco, catalog# 11875-093) supplemented with 10% fetal bovine serum (FBS), 6.25 mM HEPES, 2.5 mM L-glutamate, 0.625 mM sodium pyruvate, 0.625× nonessential amino acid solution, 62.5 μM 2-mercaptoethanol, penicillin (125 U/ml), streptomycin (125 μg/ml), and gentamycin (6.25 μg/ml) unless stated otherwise.

Antibodies

Antibodies used in this study are listed in **Table 1**. Anti-hPD-1 mAbs, HM266 and J116, and anti-mouse PD-L1 mAb, 1-111A, were prepared from the culture supernatants of hybridoma cells. The hybridoma cells were grown in CD hybridoma medium (Gibco, catalog# 11279-023) supplemented with 8 mM L-glutamine, penicillin (20 U/ml) and streptomycin (20 μg/ml) using a CELLline bioreactor flask (Duran Wheaton Kimble, catalog# WCL1000). Antibodies in the culture supernatant were purified with Protein A-R28 (Ab-Capcher ExTra; Protenova, catalog# P-003).

Purification of naïve CD4⁺ cells

Naïve CD4⁺ T cells were sorted from DO11.10 splenocytes. After the lysis of red blood cells using ACK lysing buffer (Gibco, catalog# A10492-01), spleen cells were stained using FITC-labeled antibodies to CD8, CD19, CD45R/B220, CD11c, CD11b, CD49b and I-A/I-E. Antibody-labeled cells

Antibody	Clone	Source	Catalog Number	RRID
Ultra-LEAF™ Purified anti-mouse CD3ε Antibody	145-2C11	Biolegend	100340	AB_11149115
FITC anti-mouse CD4 Antibody	RM4-5	Biolegend	100510	AB_312713
PerCP/Cyanine5.5 anti-mouse CD4 Antibody	GK1.5	Biolegend	100434	AB_893324
Brilliant Violet 510™ anti-mouse CD4 Antibody	GK1.5	Biolegend	100449	AB_2564587
Brilliant Violet 605™ anti-mouse CD4 Antibody	RM4-5	Biolegend	100451	AB_2564591
FITC anti-mouse CD8a Antibody	53-6.7	Biolegend	100706	AB_312745
FITC anti-mouse/human CD11b Antibody	M1/70	Biolegend	101206	AB_312789
FITC anti-mouse CD11c Antibody	N418	Biolegend	117306	AB_313775
TruStain FcX™ (anti-mouse CD16/32) Antibody	93	Biolegend	101320	AB_1574975
FITC anti-mouse CD19 Antibody	6D5	Biolegend	115506	AB_313641
FITC anti-mouse/human CD45R/B220 Antibody	RA3-6B2	Biolegend	103206	AB_312991
FITC anti-mouse CD49b (pan-NK cells) Antibody	DX5	Biolegend	108906	AB_313413
PE anti-mouse CD62L Antibody	MEL-14	Biolegend	104408	AB_313095
APC anti-mouse CD62L Antibody	MEL-14	Biolegend	104412	AB_313099
Brilliant Violet 421™ anti-rat CD90/mouse CD90.1 (Thy-1.1) Antibody	OX-7	Biolegend	202529	AB_10899572
Brilliant Violet 421™ anti-mouse CD279 (PD-1) Antibody	29F.1A12	Biolegend	135218	AB_2561447
Brilliant Violet 711™ anti-human CD279 (PD-1) Antibody	EH12.2H7	Biolegend	329928	AB_2562911
PE anti-human CD279 (PD-1) Antibody	EH12.2H7	Biolegend	329906	AB_940483
Purified anti-human PD-1 antibody	HM266	-	-	-
Purified anti-human PD-1 antibody	J116	-	-	-
Purified anti-mouse PD-L1 antibody	1-111A	-	-	-
FITC anti-mouse I-A/I-E Antibody	M5/114.15.2	Biolegend	107606	AB_313321
Brilliant Violet 510™ anti-mouse TCR β chain Antibody	H57-597	Biolegend	109234	AB_2562350
Alexa Fluor647 anti-SiglecF	E50-2440	BD Biosciences	562680	AB_2687570
Brilliant Violet 421™ anti-T-bet Antibody	4B10	Biolegend	644816	AB_10959653
Gata-3 Monoclonal Antibody (TWAJ), PE, eBioscience™	TWAJ	eBioscience	12-9966-42	AB_1963600
Purified Rat IgG2a, κ Isotype Ctrl Antibody	RTK2758	BioXCell	400502	AB_326523
InVivoMAb mouse IgG1 isotype	MOPC-21	BioXCell	BE0083	AB_1107784

Table 1.

List of antibodies used in this study

control				
InVivoMAb anti-mouse IFN γ	XMG1.2	BioXCell	BE0055	AB_1107694
Brilliant Violet 650™ anti-mouse IFN- γ Antibody	XMG1.2	Biolegend	505832	AB_2734492
InVivoMAb anti-mouse IL-4	11B11	BioXCell	BE0045	AB_1107707
PE/Cyanine7 anti-mouse IL-4 Antibody	11B11	Biolegend	504118	AB_10898116
PE anti-mouse/human IL-5 Antibody	TRFK5	Biolegend	504304	AB_315328
IL-13 Monoclonal Antibody (eBio13A), eFluor™ 660, eBioscience™	eBio13A	eBioscience	50-7133-82	AB_2574279
InVivoMAb anti-mouse IL-12 p40	C17.8	BioXCell	BE0051	AB_1107698

Table 1. (continued)

were depleted by labeling with anti-FITC MACS beads (Miltenyi Biotec, catalog# 130-048-701) and subsequent negative sorting using autoMACS Pro Separator (Miltenyi Biotec). The enriched CD4⁺ T cell fraction was further stained with PerCP-Cy5.5-labeled anti-CD4 mAb and APC-labeled anti-CD62L mAb, and CD4⁺ CD62L⁺ cells were purified using FACS Aria or FACSMelody cell sorter (BD Biosciences).

PD-1 expression

Purified DO11.10 CD4⁺ CD62L⁺ cells were stimulated with OVA₃₂₃₋₃₃₉ peptide (Eurofins) using the same number of parental or PD-L1-deficient A20 cells as APC. The cell mixture was added in a 96-well flat-bottom plate, briefly spun down, and cultured in a CO₂ incubator for the indicated period of time. PD-1 expression was analyzed by flow cytometry using BV421-anti-mouse PD-1 mAb or BV711-anti-hPD-1 mAb. Flow cytometry was performed by FACSFortessa X-20 (BD Biosciences) and analyzed with FlowJo software (BD Biosciences).

T cell proliferation

Purified DO11.10 CD4⁺ CD62L⁺ T cells were labeled with CFSE (Invitrogen) in PBS at 5 μM for 30 min at 37 °C, followed by three rounds of washing by culture media prior to coculture with APCs. CFSE-labeled T cells were stimulated with OVA₃₂₃₋₃₃₉ peptide in the presence of A20 cells. Time-dependent changes of CFSE signals of CD4⁺ cells were analyzed by flow cytometry.

T helper subset differentiation

CD4⁺ CD62L⁺ T cells from DO11.10 mice were stimulated with OVA₃₂₃₋₃₃₉ peptide using the same number of parental or PD-L1-deficient A20 cells. Prior to the co-culture, A20 cells were pretreated with 0.5 mg/mL mitomycin C (Fujifilm Wako Chemicals, catalog# 139-18711) at 37 °C for 2 hours and used after extensive wash. All functional differentiation experiments were conducted in the presence of IL-2 (2 ng/mL; Peprotech, catalog# 212-12). Additional cytokines or anti-cytokine antibodies were used for the specific skewing into Th1 or Th2 cells: IL-12p70 (1 ng/mL; Peprotech, catalog# 210-12), IFN-γ (1 ng/mL; Peprotech, catalog# 315-05) and anti-IL-4 mAb (5 μg/mL) for Th1; IL-4 (1 ng/mL; Peprotech, catalog# 214-14), anti-IFN-γ mAb (5 μg/mL), anti-IL-12 p40 mAb (5 μg/mL) for Th2. After 3 days, activated CD4⁺ T cells were restimulated with OVA₃₂₃₋₃₃₉ in the same conditions for 1 or 2 more days. For the PD-L1 blockade experiment, anti-mouse PD-L1 mAb (5 μg/mL) or its isotype control mAb (RTK2758) was included in the culture. The effect of anti-hPD-1 agonist mAb, HM266 and J116 (5 μg/mL), was compared with that of isotype control mAb (MOPC21).

Cytokine production

To evaluate cytokine-producing activity, DO11.10 cells after functional differentiation were washed with the cell culture media. The same number of activated CD4⁺ T cells (2 × 10⁵ cells) were restimulated with plate-bound anti-CD3 mAb on 96-well flat-bottom plates for 24 h. Cytokine levels in the culture supernatant were quantified by the use of Mouse IFN-γ DuoSet ELISA (R&D Systems, catalog# DY485), Mouse IL-4 DuoSet ELISA (R&D Systems, catalog# DY404) and Mouse IL-13 DuoSet ELISA (R&D Systems, catalog# DY413).

Intracellular staining

Cells were stimulated with plate-bound anti-CD3 mAb on 96-well flat-bottom plates for 6 h in the presence of BD GolgiPlug (BD Biosciences, catalog# 555029) for the final 2 hours of culture. The stimulated cells were treated with anti-mouse CD16/32 mAb, followed by Fixable viability dye eFluor780 (eBioscience, catalog# 65-0865-14) and BV510-anti-CD4 mAb. Cell surface staining may include BV421-anti-Thy1.1 mAb when necessary. After fixation and permeabilization, cells were stained with BV650-anti-IFN-γ mAb, PE-anti-IL-5 mAb, PE-Cy7-anti-IL-4 mAb and eFluor 660-anti-IL-13 mAb. For the detection of transcription factors, cells stained with BV510-anti-TCRβ mAb and

BV605-anti-CD4 mAb were fixed and permeabilized using True-Nuclear Transcription Factor Buffer Set (BioLegend, catalog# 424401) before staining with BV421-anti-T-bet mAb and PE-anti-GATA3 mAb.

Retroviral transduction

hPD-1 retroviral plasmid was generated by inserting hPD-1 cDNA into MSCV-IRES-Thy1.1 DEST (Addgene, catalog# 17442). Point mutations were introduced by using QuikChange II site-directed mutagenesis kit (Agilent, catalog# 200523). For bicistronic transduction of hPD1/GFP fusion and mSHP2-C459S protein, IRES-Thy1.1 sequence of MSCV-IRES-Thy1.1 was replaced with hPD-1/GFP-P2A-mSHP2-C459S sequence. The retroviral plasmids were transfected into Plat-E cells (Cell Biolabs, RRID: CVCL_B488) using Fugene HD Transfection Reagent (Promega, catalog# E2311). After 2-3 days, retroviral supernatant was pooled and passed through 0.45 μ m Minisart Syringe Filter (Sartorius, catalog# 16533). The retrovirus supernatant was added to Retronectin (Takara Bio, catalog# T100A)-coated culture plate and centrifuged at 32 °C for 2 h. Purified naïve DO11.10 CD4⁺ T cells as described above were stimulated using Dynabeads Mouse T-Activator CD3/CD28 (Gibco, catalog# 11452D) for 24 h. Activated T cells were placed on retrovirus-coated culture plate and centrifuged at 800 x g for 10 min at 32 °C. This procedure was repeated on the next day, and hPD-1-transduced T cells was used on day 3 for further studies. Cellular expression of hPD-1 was confirmed by flow cytometry using PE-anti-hPD-1 mAb.

Agonistic activity of anti-hPD-1 mAbs

hPD-1-expressing DO11.10 T cell hybridoma cells (5×10^4 cells) were stimulated with OVA₃₂₃₋₃₃₉ (2 μ g/ml) presented on hPD-L1⁺ mFc γ RIIB⁺ IIA1.6 B lymphoma cells (1×10^4 cells). Culture supernatants were collected after 18 h, and IL-2 levels were determined using mouse IL-2 DuoSet ELISA (R&D Systems, catalog# DY402).

Sensitization of BMDCs with HDM extract

Bone marrow cells from the femurs were treated with ACK lysis buffer and resuspended in the cell culture media. After incubating in a culture plate for 4 h at 37 °C, loosely adherent cells were collected and further cultured in media supplemented with recombinant mouse GM-CSF (20 ng/mL; Peprotech, catalog# 315-03). The same fresh GM-CSF-supplemented media was supplied on day 2 and 4. On day 6, the cells were further incubated in media without GM-CSF for 1 day before using them as BMDCs. HDM extract (*Dermatophagoides pteronyssinus*; Greer Laboratories, catalog# B70) was given to cultured BMDCs at 1 μ g (total protein)/ml, and the supernatant was collected after 24 h.

HDM extract-induced allergy model

hPD-1-KI C57BL/6 mice were randomly assigned to groups and were sensitized with HDM extract (10 ng Der p 1/mouse i.p.). A week later, intranasal challenge with HDM extract (25 μ g of whole protein/25 μ L per mouse) started and repeated for 7 consecutive days. The mice were euthanized 4 h after the last challenge. The BAL fluid was collected by introducing 1 ml PBS through the trachea. The lungs were processed with Lung Dissociation Kit (Miltenyi Biotec, catalog# 130-095-927) and GentleMACS Dissociator (Miltenyi Biotec) according to the manufacturer's instructions, and the lung mononuclear cell were isolated by density centrifugation using 40 and 70% Percoll (Cytiva, catalog# 17089102). For the detection of eosinophils, BAL cells were incubated with anti-mouse CD16/32 mAb to block Fc receptors, and stained with FITC-anti-CD11c mAb, Alexa Fluor 647-anti-SiglecF mAb and Fixable viability dye eFluor780. Eosinophils were defined as CD11c⁺ SiglecF⁺ population by flow cytometry. For intracellular cytokine staining, BAL cells were stimulated with plate-bound anti-CD3 mAb on 96-well flat-bottom plates for 6 h in the presence of BD GolgiPlug for the final 2 hours of culture and were stained as described above.

Histology

Lungs from HDM-induced allergic mice were collected and fixed in 10% neutral-buffered formalin. Fixed lungs were then processed for PAS staining by Applied Medical Research Laboratory.

Plasma IgE levels

Levels of total IgE and HDM-specific IgE in the plasma were determined by using ELISA MAX™ Deluxe Set Mouse IgE (Biolegend, catalog# 432404) and Mouse Serum Anti-HDM Dermatophagoides pteronyssinus IgE Antibody Assay Kit (Chondrex, catalog# 3037) according to manufacturer's instructions.

Statistics

Data are expressed as mean $\pm$ SEM of biological replicates. Data shown in Figures are representative of 2-3 independent experiments with similar results. Statistical significance was calculated by two-tailed Student's t-test for two-group comparison and Tukey-Kramer test for multiple comparison. P values less than 0.05 were considered significant.

Data Availability

The data are available from the corresponding author upon reasonable request.

Author Contributions

Conceptualization: MT, TH, AO

Methodology: MT, AO

Investigation: MT, NI, YN, KS, YT, PL, AO

Resources (hPD-1-knock-in mice): HK

Writing – Original Draft: MT, AO

Writing – Review and Editing: MT, NI, YN, TH, AO

Supervision: TH, AO

Funding Acquisition: TH

Conflicts of Interest

M.T., K.S., Y.T., T.H. and A.O. are inventors on patent applications (WO2021/241523A1, WO2022/239820) submitted by Foundation for Biomedical Research and Innovation at Kobe, National Institutes of Biomedical Innovation, Health and Nutrition, and Meiji Seika Pharma Co.

Ltd. that cover anti-human PD-1 agonist antibodies and their therapeutic application. K.S. and Y.T. are employees of Meiji Seika Pharma Co. Ltd. The remaining authors declare no competing interests.

Acknowledgements

This work was supported by Meiji Seika Pharma Co. Ltd. and the Foundation for Biomedical Research and Innovation at Kobe. The authors thank to Miwa Okada, Kumiko Yonezaki, Yukako Kamita and Chiyomi Ito for the antibody production and mouse colony management. Anti-hPD-1 agonist antibodies were previously developed in our collaboration with Drs. Satoshi Nagata and Haruhiko Kamada (National Institutes of Biomedical Innovation, Health and Nutrition).

References

1. Nakayama T., Hirahara K., Onodera A., Endo Y., Hosokawa H., Shinoda K., Tumes D. J., Okamoto Y. (2017) **Th2 Cells in Health and Disease** *Annu Rev Immunol* **35**:53–84
2. Lloyd C. M., Snelgrove R. J. (2018) **Type 2 immunity: Expanding our view** *Sci Immunol* **3**
3. Gowthaman U. *et al.* (2019) **Identification of a T follicular helper cell subset that drives anaphylactic IgE** *Science* **365**
4. Halim T. Y., Steer C. A., Matha L., Gold M. J., Martinez-Gonzalez I., McNagny K. M., McKenzie A. N., Takei F. (2014) **Group 2 innate lymphoid cells are critical for the initiation of adaptive T helper 2 cell-mediated allergic lung inflammation** *Immunity* **40**:425–435
5. Oliphant C. J. *et al.* (2014) **MHCII-mediated dialog between group 2 innate lymphoid cells and CD4(+) T cells potentiates type 2 immunity and promotes parasitic helminth expulsion** *Immunity* **41**:283–295
6. Van Dyken S. J. *et al.* (2016) **A tissue checkpoint regulates type 2 immunity** *Nat Immunol* **17**:1381–1387
7. Morel P. A. (2018) **Differential T-cell receptor signals for T helper cell programming** *Immunology* **155**:63–71
8. Walker J. A., McKenzie A. N. J. (2018) **T(H)2 cell development and function** *Nat Rev Immunol* **18**:121–133
9. El-Naccache D. W., Hasko G., Gause W. C. (2021) **Early Events Triggering the Initiation of a Type 2 Immune Response** *Trends Immunol* **42**:151–164
10. Chemnitz J. M., Parry R. V., Nichols K. E., June C. H., Riley J. L. (2004) **SHP-1 and SHP-2 associate with immunoreceptor tyrosine-based switch motif of programmed death 1 upon primary human T cell stimulation, but only receptor ligation prevents T cell activation** *J Immunol* **173**:945–954
11. Yokosuka T., Takamatsu M., Kobayashi-Imanishi W., Hashimoto-Tane A., Azuma M., Saito T. (2012) **Programmed cell death 1 forms negative costimulatory microclusters that directly inhibit T cell receptor signaling by recruiting phosphatase SHP2** *J Exp Med* **209**:1201–1217
12. Okazaki T., Chikuma S., Iwai Y., Fagarasan S., Honjo T. (2013) **A rheostat for immune responses: the unique properties of PD-1 and their advantages for clinical application** *Nat Immunol* **14**:1212–1218
13. Topalian S. L. *et al.* (2012) **Safety, activity, and immune correlates of anti-PD-1 antibody in cancer** *N Engl J Med* **366**:2443–2454
14. Pauken K. E., Torchia J. A., Chaudhri A., Sharpe A. H., Freeman G. J. (2021) **Emerging concepts in PD-1 checkpoint biology** *Semin Immunol* **52**

15. Chamoto K., Yaguchi T., Tajima M., Honjo T. (2023) **Insights from a 30-year journey: function, regulation and therapeutic modulation of PD1** *Nat Rev Immunol* **23**:682–695
16. Weide B. *et al.* (2016) **Baseline Biomarkers for Outcome of Melanoma Patients Treated with Pembrolizumab** *Clin Cancer Res* **22**:5487–5496
17. Bernard-Tessier A., Jeanville P., Champiat S., Lazarovici J., Voisin A. L., Mateus C., Lambotte O., Annereau M., Michot J. M. (2017) **Immune-related eosinophilia induced by anti-programmed death 1 or death-ligand 1 antibodies** *Eur J Cancer* **81**:135–137
18. Mirza S., Hill E., Ludlow S. P., Nanjappa S. (2017) **Checkpoint inhibitor-associated drug reaction with eosinophilia and systemic symptom syndrome** *Melanoma Res* **27**:271–273
19. Jodai T. *et al.* (2019) **A potential mechanism of the onset of acute eosinophilic pneumonia triggered by an anti-PD-1 immune checkpoint antibody in a lung cancer patient** *Immun Inflamm Dis* **7**:3–6
20. Fukushima A., Yamaguchi T., Azuma M., Yagita H., Ueno H. (2006) **Involvement of programmed death-ligand 2 (PD-L2) in the development of experimental allergic conjunctivitis in mice** *Br J Ophthalmol* **90**:1040–1045
21. Tanaka R. *et al.* (2022) **Differential Involvement of Programmed Cell Death Ligands in Skin Immune Responses** *J Invest Dermatol* **142**:145–154
22. Matsumoto K. *et al.* (2004) **B7-DC regulates asthmatic response by an IFN-gamma-dependent mechanism** *J Immunol* **172**:2530–2541
23. Lewkowich I. P., Lajoie S., Stoffers S. L., Suzuki Y., Richgels P. K., Dienger K., Sproles A. A., Yagita H., Hamid Q., Wills-Karp M. (2013) **PD-L2 modulates asthma severity by directly decreasing dendritic cell IL-12 production** *Mucosal Immunol* **6**:728–739
24. Akbari O., Stock P., Singh A. K., Lombardi V., Lee W. L., Freeman G. J., Sharpe A. H., Umetsu D. T., Dekruyff R. H. (2010) **PD-L1 and PD-L2 modulate airway inflammation and iNKT-cell-dependent airway hyperreactivity in opposing directions** *Mucosal Immunol* **3**:81–91
25. Suzuki K., Tajima M., Tokumaru Y., Oshiro Y., Nagata S., Kamada H., Kihara M., Nakano K., Honjo T., Ohta A. (2023) **Anti-PD-1 antibodies recognizing the membrane-proximal region are PD-1 agonists that can down-regulate inflammatory diseases** *Sci Immunol* **8**
26. Ruterbusch M., Pruner K. B., Shehata L., Pepper M. (2020) **In Vivo CD4(+) T Cell Differentiation and Function: Revisiting the Th1/Th2 Paradigm** *Annu Rev Immunol* **38**:705–725
27. Yamane H., Paul W. E. (2013) **Early signaling events that underlie fate decisions of naive CD4(+) T cells toward distinct T-helper cell subsets** *Immunol Rev* **252**:12–23
28. Gregory L. G., Lloyd C. M. (2011) **Orchestrating house dust mite-associated allergy in the lung** *Trends Immunol* **32**:402–411
29. Liang S. C., Greenwald R. J., Latchman Y. E., Rosas L., Satoskar A., Freeman G. J., Sharpe A. H. (2006) **PD-L1 and PD-L2 have distinct roles in regulating host immunity to cutaneous leishmaniasis** *Eur J Immunol* **36**:58–64

30. Lu L. *et al.* (2022) **Excessive immunosuppression by regulatory T cells antagonizes T cell response to schistosome infection in PD-1-deficient mice** *PLoS Pathog* **18**
31. Iwai Y., Ishida M., Tanaka Y., Okazaki T., Honjo T., Minato N. (2002) **Involvement of PD-L1 on tumor cells in the escape from host immune system and tumor immunotherapy by PD-L1 blockade** *Proc Natl Acad Sci U S A* **99**:12293–12297
32. Blank C., Brown I., Peterson A. C., Spiotto M., Iwai Y., Honjo T., Gajewski T. F. (2004) **PD-L1/B7H-1 inhibits the effector phase of tumor rejection by T cell receptor (TCR) transgenic CD8+ T cells** *Cancer Res* **64**:1140–1145
33. Iwai Y., Terawaki S., Honjo T. (2005) **PD-1 blockade inhibits hematogenous spread of poorly immunogenic tumor cells by enhanced recruitment of effector T cells** *Int Immunol* **17**:133–144
34. Ashoori M. D., Suzuki K., Tokumaru Y., Ikuta N., Tajima M., Honjo T., Ohta A. (2020) **Inactivation of the PD-1-Dependent Immunoregulation in Mice Exacerbates Contact Hypersensitivity Resembling Immune-Related Adverse Events** *Front Immunol* **11**
35. Hirano T. *et al.* (2021) **PD-L1 on mast cells suppresses effector CD8(+) T-cell activation in the skin in murine contact hypersensitivity** *J Allergy Clin Immunol* **148**:563–573
36. Zhou S. *et al.* (2016) **Blockade of PD-1 Signaling Enhances Th2 Cell Responses and Aggravates Liver Immunopathology in Mice with Schistosomiasis japonica** *PLoS Negl Trop Dis* **10**
37. Yamazaki T. *et al.* (2002) **Expression of programmed death 1 ligands by murine T cells and APC** *J Immunol* **169**:5538–5545
38. Loke P., Allison J. P. (2003) **PD-L1 and PD-L2 are differentially regulated by Th1 and Th2 cells** *Proc Natl Acad Sci U S A* **100**:5336–5341
39. Nie X. *et al.* (2018) **B7-DC (PD-L2) costimulation of CD4(+) T-helper 1 response via RGMb** *Cell Mol Immunol* **15**:888–897
40. Okazaki T. *et al.* (2011) **PD-1 and LAG-3 inhibitory co-receptors act synergistically to prevent autoimmunity in mice** *J Exp Med* **208**:395–407
41. Yang J., Lagana S. M., Saenger Y. M., Carvajal R. D. (2019) **Dual checkpoint inhibitor-associated eosinophilic enteritis** *J Immunother Cancer* **7**
42. Scanvion Q. *et al.* (2020) **Moderate-to-severe eosinophilia induced by treatment with immune checkpoint inhibitors: 37 cases from a national reference center for hypereosinophilic syndromes and the French pharmacovigilance database** *Oncoimmunology* **9**
43. Simon S. C. S., Hu X., Panten J., Grees M., Renders S., Thomas D., Weber R., Schulze T. J., Utikal J., Umansky V. (2020) **Eosinophil accumulation predicts response to melanoma treatment with immune checkpoint inhibitors** *Oncoimmunology* **9**
44. Paul W. E. (2010) **What determines Th2 differentiation, in vitro and in vivo?** *Immunol Cell Biol* **88**:236–239

45. Cote-Sierra J., Foucras G., Guo L., Chiodetti L., Young H. A., Hu-Li J., Zhu J., Paul W. E. (2004) **Interleukin 2 plays a central role in Th2 differentiation** *Proc Natl Acad Sci U S A* **101**:3880–3885
46. Yamane H., Zhu J., Paul W. E. (2005) **Independent roles for IL-2 and GATA-3 in stimulating naive CD4⁺ T cells to generate a Th2-inducing cytokine environment** *J Exp Med* **202**:793–804
47. Chikuma S., Terawaki S., Hayashi T., Nabeshima R., Yoshida T., Shibayama S., Okazaki T., Honjo T. (2009) **PD-1-mediated suppression of IL-2 production induces CD8⁺ T cell anergy in vivo** *J Immunol* **182**:6682–6689
48. Constant S., Pfeiffer C., Woodard A., Pasqualini T., Bottomly K. (1995) **Extent of T cell receptor ligation can determine the functional differentiation of naive CD4⁺ T cells** *J Exp Med* **182**:1591–1596
49. Pfeiffer C., Stein J., Southwood S., Ketelaar H., Sette A., Bottomly K. (1995) **Altered peptide ligands can control CD4 T lymphocyte differentiation in vivo** *J Exp Med* **181**:1569–1574
50. van Panhuys N., Klauschen F., Germain R. N. (2014) **T-cell-receptor-dependent signal intensity dominantly controls CD4(+) T cell polarization In Vivo** *Immunity* **41**:63–74
51. Hosken N. A., Shibuya K., Heath A. W., Murphy K. M., O'Garra A. (1995) **The effect of antigen dose on CD4⁺ T helper cell phenotype development in a T cell receptor-alpha beta-transgenic model** *J Exp Med* **182**:1579–1584
52. Ismail N., Bretscher P. A. (1999) **The Th1/Th2 nature of concurrent immune responses to unrelated antigens can be independent** *J Immunol* **163**:4842–4850
53. McAlees J. W. *et al.* (2015) **Differential control of CD4(+) T-cell subsets by the PD-1/PD-L1 axis in a mouse model of allergic asthma** *Eur J Immunol* **45**:1019–1029
54. Biemond M., Vremec D., Gray D. H., Hodgkin P. D., Heinzel S. (2024) **Programmed death receptor 1 (PD-1) ligand Fc fusion proteins reduce T-cell proliferation in vitro independently of PD-1** *Immunol Cell Biol* **102**:117–130
55. Yamashita M., Kimura M., Kubo M., Shimizu C., Tada T., Perlmutter R. M., Nakayama T. (1999) **T cell antigen receptor-mediated activation of the Ras/mitogen-activated protein kinase pathway controls interleukin 4 receptor function and type-2 helper T cell differentiation** *Proc Natl Acad Sci U S A* **96**:1024–1029
56. Goplen N., Karim Z., Guo L., Zhuang Y., Huang H., Gorska M. M., Gelfand E., Pages G., Pouyssegur J., Alam R. (2012) **ERK1 is important for Th2 differentiation and development of experimental asthma** *FASEB J* **26**:1934–1945
57. Tripathi P., Sahoo N., Ullah U., Kallionpaa H., Suneja A., Lahesmaa R., Rao K. V. (2012) **A novel mechanism for ERK-dependent regulation of IL4 transcription during human Th2-cell differentiation** *Immunol Cell Biol* **90**:676–687
58. Tao X., Constant S., Jorritsma P., Bottomly K. (1997) **Strength of TCR signal determines the costimulatory requirements for Th1 and Th2 CD4⁺ T cell differentiation** *J Immunol* **159**:5956–5963
59. Helou D. G. *et al.* (2020) **PD-1 pathway regulates ILC2 metabolism and PD-1 agonist treatment ameliorates airway hyperreactivity** *Nat Commun* **11**

60. Helou D. G. *et al.* (2023) **Human PD-1 agonist treatment alleviates neutrophilic asthma by reprogramming T cells** *J Allergy Clin Immunol* **151**:526–538

Editors

Reviewing Editor

Ana Maria Faria

Universidade Federal de Minas Gerais, Belo Horizonte, Brazil

Senior Editor

Carla Rothlin

Yale University, New Haven, United States of America

Reviewer #1 (Public review):

Summary:

The authors analyze the roles of PD-1 in the early stages (pre-activation) of T cell differentiation and show that naïve CD4⁺ T cell differentiation is altered, especially Th2 differentiation is strongly impaired, upon early PD-1 stimulation. The results have important implications for the immunotherapy area, but I think the manuscript requires some revisions.

Strengths:

(1) Novel Insights into PD-1 in Early T Cell Differentiation:

The study provides new insights into the role of PD-1 during the pre-activation phase of T cell differentiation, particularly its impact on naïve CD4⁺ T cells and Th2 differentiation. This is a significant contribution to immunotherapy research.

(2) Relevance to Immunotherapy:

The findings have potential implications for the development of immunotherapies by demonstrating how PD-1 signaling affects specific T cell subsets early in differentiation.

Weaknesses:

(1) Inconsistent and Confusing Data:

There are contradictions between the figures and the conclusions, particularly regarding IL-4 and IFN γ production in PD-1-expressing cells. This raises concerns about data interpretation and experimental accuracy.

(2) Unclear Experimental Rationale:

The reviewer questions the rationale behind key methodological choices, such as the high concentration of PDL-1 antibody and varying OVA peptide concentrations. These decisions need more justification.

<https://doi.org/10.7554/eLife.100879.1.sa1>

Reviewer #2 (Public review):

Summary:

The authors try to demonstrate that PD-1 regulates not only the quantity but also the quality of the immune response determining the Th differentiation. The authors suggest that the

ability of PD-1 agonists to dampen Th2 differentiation could be exploited in allergies or classical Th2-mediated disease as a therapeutical approach.

Strengths:

The authors performed a series of elegant experiments using OVA-specific CD4 T cells from mice, showing a strong reduction of Th2 differentiation *in vitro*. They also perform some experiments with a model of allergies, showing an amelioration of the phenotype after administration of PD-1 agonist with a reduction of Th2 cells.

Weaknesses:

The authors perform all the experiments using DO11.10 mouse cells. Such cells have a TCR with very high affinity, it would be relevant to repeat at least some of the *in vitro* assays in a more physiological setting (you can immunise mice with ova to increase the pool of OVA-specific T cells, and then repeat the restimulation experiment). Also, a longer kinetic would be of interest to see the effect of the agonist on Th1 cells.

Another drawback is the lack of experiments with human cells. It would be really important to repeat the experiments with CD4 T cells from healthy donors (the antibody that the authors use as PD-1 agonist is human, so it would not be a complicated experiment).

It would be also interesting to show in the allergic disease model the effect of the agonism on the T cell response in general.

<https://doi.org/10.7554/eLife.100879.1.sa0>

Author Response:

We are grateful to the reviewers for their encouraging comments and constructive suggestions. These suggestions will be valuable to improve the revised manuscript.

Reviewer 1:

PD-1 signaling is suppressive to the establishment of cytokine-producing effector cells in general. However, as the reviewer pointed out, one of the results in Fig. 2H showing a decrease of IFN-gamma-producing cells is against this trend. The data indicate percentages of cytokine-producing cells, which are not always consistent with the absolute number of activated T cells. Nonetheless, we plan additional experiments in order to address the question.

For PD-1YFYF experiments in Figs. 3-5, there were moderate changes in cytokine production between wild-type and mutant PD-1. We conducted gene transduction to newly prepared T cells in each experiment. In addition, to monitor the immunosuppressive effect of PD-1 agonist antibodies, these T cells were stimulated using PD-L1-deficient APC. Therefore, we think these cytokine levels were most likely a technical variation, but not specific function of PD-1YFYF.

Anti-PD-L1 mAb was used for the optimal blockade of PD-1/PD-L1 blockade, and the concentration of antibody (5 microg/ml) is within a normal range for this purpose. We used variable concentrations of OVA peptide to set up experiments with different intensities of TCR stimulation. TCR signal intensity has been shown to affect CD4⁺ T cell differentiation into Th1 and Th2 cells. We lowered the peptide concentration to test the effect of PD-1 signals under the suboptimal TCR stimulation.

Reviewer 2:

Antigen-specific T cells from immunized mice are not ideal for Th differentiation studies because activated T cells in response to the antigen might have already undergone functional differentiation in vivo. Incorporating the reviewer's suggestion, we will test alternative approach including human CD4⁺ T cells.

For the allergy model, we will expand the analysis for inflammatory effectors.

<https://doi.org/10.7554/eLife.100879.1.sa3>